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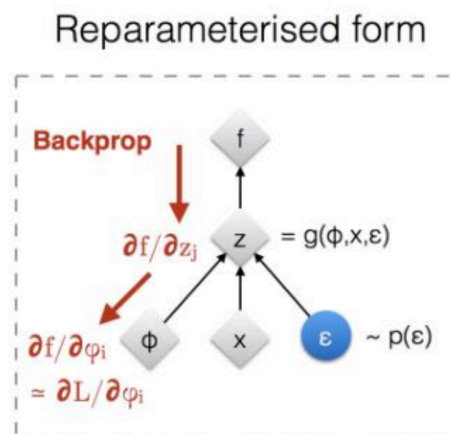
Pen & Paper Exercises

1. Reparametrization trick: Explain how the trick works and what problem it addresses.

The reparameterization trick allows gradients to flow through random sampling by separating the source of randomness from the network's parameters. Instead of sampling directly, it expresses the sample as a deterministic transformation of a fixed noise variable, making backpropagation possible. To shorten sampling operation is non-differentiable $z \sim \mathcal{N}(\mu, \sigma)$ when we don't have epsilon.

- Fixed μ and σ scaled by random constants drawn from a prior distribution

- $z = \mu + \sigma \odot \epsilon$
- $\epsilon \sim \mathcal{N}(0,1)$



2. Explain in your own words why optimization is hard in the case of GANs. What are common ways to improve GAN training stability?

Optimization in GANs is difficult because the objective is non-convex and involves finding a balance between two competing networks, which can lead to issues like mode collapse, vanishing gradients, or oscillations in training. To improve stability, common strategies include using Wasserstein loss with gradient penalty, spectral normalization, or techniques like

minibatch discrimination to prevent mode collapse. Additionally, carefully tuning the learning rate and applying label smoothing can help stabilize the training process.

GAN limitations

- Hard to find optima
 - Optimization is a min-max game and it is hard to achieve Nash equilibrium
- Vanishing gradient
 - When the discriminator is perfect, the loss falls to 0 with no gradient to update the loss
- Mode collapse
 - Generator might collapse to a setting where it generates the same output

3. Shortly compare GANs and VAEs. In which setting would you select one over the other?

GANs create realistic data by having a generator and discriminator compete, while VAEs focus on learning the structure of data by modeling probabilities and generating continuous outputs. GANs are better for generating highly realistic images, while VAEs are better when you need smooth data representations or want to explore variations in data.

